William (Songyang) Wang

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Machine Learning Engineer | **Ph.D.** Developing machine learning algorithms and infrastructure for deep learning model training. Python and PyTorch, distributed training (DDP), GPU throughput optimization, model deployment, terabyte-scale data.

Education

Boston University — Boston, MA
Ph.D. in Biomedical Engineering | *GPA: 3.88* | ***AI Certificate**: Deep Learning Specialization* — *Aug 2020 – May 2026*

University of California, Irvine — Irvine, CA
B.S. in Biomedical Engineering & Economics | *GPA: 3.93* | ***Summa Cum Laude*** — *Aug 2016 – May 2020*

Professional Experience

Encora Therapeutics, Data Scientist Intern — Boston, MA (Remote)

Computer Vision for Clinical Tremor Severity Assessment — May 2026 – Present

- Benchmarked **six CNN architectures** with **subject-independent** CV, reaching **0.82 QWK** (manuscript in preparation).
- Matched a fully fine-tuned **134M-parameter** model by training only **∼18K parameters** on top of a frozen backbone.
- Cut model training time **11×** (10 min → 55 s/epoch) by diagnosing **GPU starvation** from sequential CPU augmentation and reconfiguring the PyTorch input pipeline for **multi-worker overlapped prefetch**.
- Deployed the model as a containerized **FastAPI** inference service on **AWS EC2**, **decoupling the API layer from model logic** so the model can be swapped and tested without touching the interface.
- Ran an **attention-module (CBAM) placement ablation** in residual CNNs, showing placement alone decides whether the model learns (inside the residual branch) or collapses to chance (after the residual sum).
- Built the PyTorch codebase as **config-driven** preprocessing, training, and evaluation modules sharing **one model definition**, and root-caused a **silent failure** (validation kappa 0.8, test 0) to a train/eval architecture mismatch.

Harvard Medical School, Visiting PhD Student — Boston, MA

Petascale EM Connectomics: Data Operations & ML Analysis — Oct 2021 – Oct 2025

- Ran data operations across the full lifecycle of a **657 TB** electron-microscopy volume: acquisition, preprocessing, **GCP** migration, and segmentation handoff to an external vendor, then built the downstream ML analysis on the returned data.
- Extended 2D stitching and 3D reconstruction on a Slurm **HPC cluster → 7× speedup**, **85% lower memory**.
- Led a **∼15-person team** (undergraduates + postdoc) with no formal authority to produce 415 proofread neuron reconstructions in ∼1.5 years, one of the largest proofread sets in a single volume, supporting a **Nature paper** (in submission).

Boston University, Graduate Research Assistant — Boston, MA

Self-Supervised Contrastive Learning for 3D Object Classification — Jun 2024 – May 2026

- **Reimplemented** the codebases of two cutting-edge AI/ML models (**SegCLR**/**SynapseCLR**) from TensorFlow in **PyTorch**, and developed a custom normalization method to bridge data domains.
- Scaled SegCLR contrastive pretraining across 2 V100 GPUs with PyTorch **DistributedDataParallel (DDP)**, learning view-invariant embeddings of neurons.
- Trained on **∼1 TB** of 3D EM volumes with **out-of-core**, per-sample loading of pre-extracted $256 \times 256 \times 256$ cutouts, and halved the batch size (128 → 64) to fit **V100 memory**, extending training to compensate.
- Shipped a **reproducible train/val downstream-classifier package** (best model **0.86 accuracy**) adopted by lab members.

Selected Projects

Brain Storm BCI Hackathon: "Compass" real-time cortical guidance tool | *2nd place* — Jan 2026

- Built the **real-time backend** for a full-stack web app guiding neurosurgical placement of a 1024-channel micro-ECoG array.
- Trained a lightweight **U-Net denoiser** (∼274 KB, <5 ms CPU inference) and a **3D CNN** decoding 2D movement-direction vectors, and added motion-aware cluster tracking for stability under array drift.
- Served **real-time inference** (<50 ms end-to-end) to the team's hands-free operating-room UI at 30–60 FPS.

Multimodal Deep Learning for Clinical Prediction | *UC Irvine, undergraduate research* — Jun 2018 – Jun 2020

- Built a hybrid **CNN-LSTM** (+3% AUC) and tuned **RF/XGBoost/LightGBM with SHAP** interpretability on multimodal data (EEG/ECG, time-series, tabular) from a **1,000+ animal** cohort, with results contributing to a funded **NIH R01**.

Teaching, Mentorship & Awards

2nd place, Brain Storm BCI Hackathon (Jan 2026). Mentored 4 seniors on a year-long senior-design project and 2 undergraduates on ML/analysis over 2 years (one defended an undergraduate honors thesis). Teaching Assistant for Engineering Mechanics (2024) and Fluid Mechanics (2023). 3 poster presentations.

Technical Skills

Proficient: Python, PyTorch, TensorFlow · Deep Learning & Model Training: Distributed Training (DDP), GPU Throughput Optimization, Out-of-Core Data Loading, Self-Supervised / Contrastive Learning (SegCLR, SupCon, SimCLR), Transfer Learning · ML Architectures: CNNs (ResNet, DenseNet, ConvNeXt, U-Net, 3D CNN), RNN/GRU · Machine Learning Infrastructure: Docker, FastAPI/REST, AWS EC2, GCP, HPC/Slurm, Git, Linux/Bash · NumPy, Pandas, Scikit-learn, MATLAB · Classical ML (XGBoost, LightGBM, CatBoost, Random Forest, SVM, GLM) · Experimental Design, Hypothesis Testing · LLM application & safety (prompt engineering, guardrails).
Familiar: Transformers · SQL, R, PySpark, Snowflake, BigQuery · WebSocket · Data Structures & Algorithms, OOP.